

DON'T PANIC

Staying Warm and Keeping Cool, Explained

START HERE: WARM THE PERSON, NOT THE HOUSE

Your furnace is gone, and the air conditioner went with it. But here's the secret the utility companies never had reason to tell you: you were never heating a house. You were heating *yourself*, wastefully, by warming every cubic foot of air in the building first.

This book works the problem from the inside out: body first, then bed, then one small room — and the whole house only if you're rich in fuel. The same thinking runs the summer in reverse: shade and night air aimed at one room and one person beat a dead central unit every time.

> **THE ONE RULE: heat and cool the smallest space that works, starting with the person.** Warming one body takes a blanket. Warming one bed takes a bottle of hot water. Warming one room takes a stove. Warming a house takes a forest. Spend like it matters, because it does.

1. THE FOUR LEAKS

Your body is a small furnace that never stops burning fuel — food — to hold itself at 98.6°. Cold is just that heat leaving faster than you make it. It leaves four ways:

- **Touch** — heat flows into anything cold you touch or lie on. The ground is the worst thief there is; it will take heat from you all night without ever warming up.
- **Wind** — moving air strips the warm layer off your skin. This is wind chill, and it's why a shell layer beats a second sweater.
- **Glow** — you radiate heat to cold surfaces around you, like a stove in reverse. A foil blanket works because it bounces the glow back.
- **Sweat** — evaporating water carries heat away. That's the whole point of sweat in summer — and a quiet killer in winter, because wet clothes keep "sweating" long after you stop.



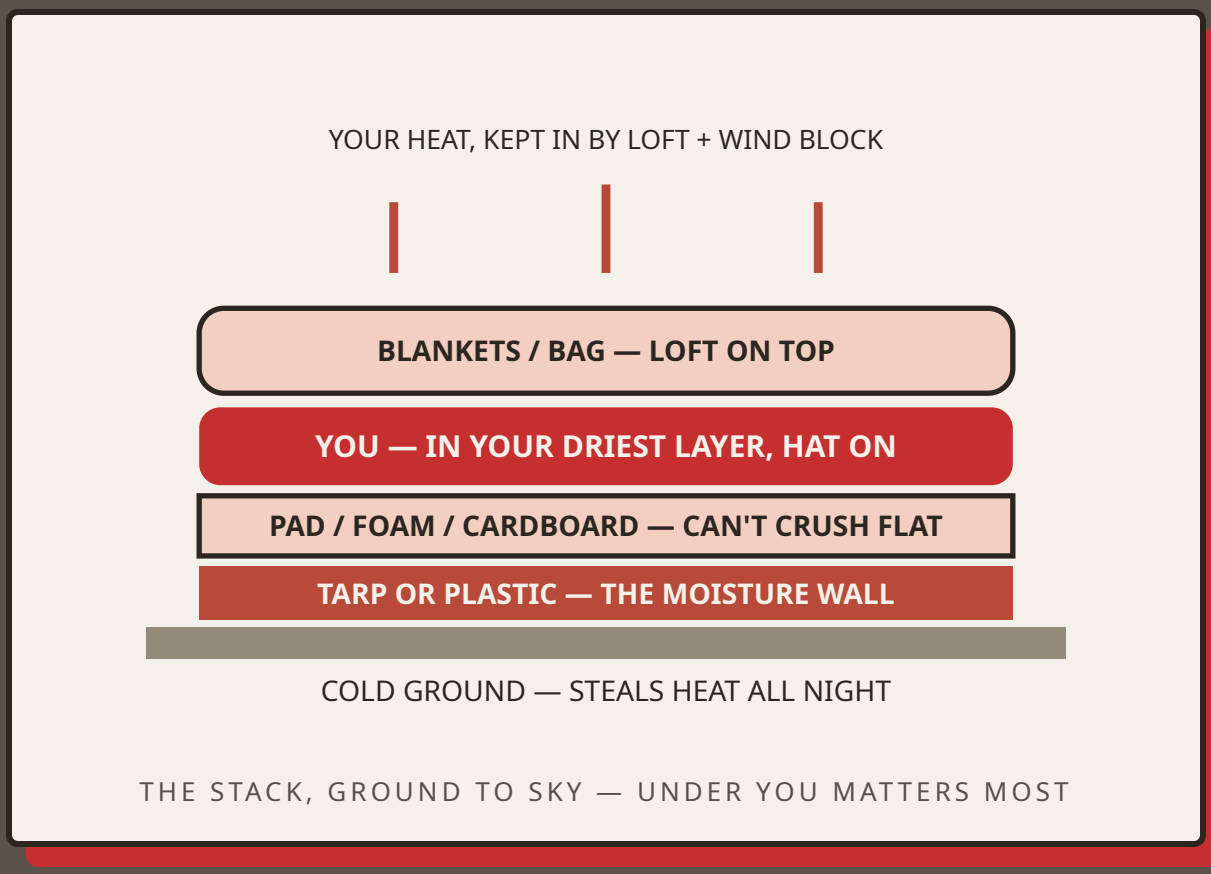
So the layer system isn't fashion, it's plumbing. **Base layer** moves sweat off your skin (never cotton if you can help it — wet cotton insulates like a wet

towel, because it is one). **Loft layer** traps still air — puffiness *is* the warmth; anything that crushes the loft crushes the warmth. **Shell layer** stops the wind from stripping it all. Change out of anything damp before you sit down, and loosen boots at night — warm feet need blood flow more than they need tight laces.

2. SLEEPING WARM: THE NIGHT IS THE TEST

You'll survive a cold day on activity. The cold night is where people get in trouble: eight still hours, body furnace turned low, ground stealing from below the whole time. Build the bed like a system:

- **Insulation under you beats blankets over you.** Your weight crushes whatever's beneath you to paper — so what goes under you must not crush: a foam pad, folded blankets, cardboard, a pile of dry leaves in a tarp. Two inches under beats four inches over.
- **Dry everything.** Change into your driest layer to sleep. Damp socks off. The day's sweat is the night's chill.
- **Eat before bed.** The furnace needs fuel to burn all night. A fatty snack at bedtime is central heating you carry inside you.
- **Hat on.** Not because heat only escapes through your head — it doesn't — but because your head is what's sticking out of the blankets.



FIELD RECIPE · SLEEP COCOON

Makes: a layered sleep system from household junk that beats one good sleeping bag.

Component	Count
Tarp or plastic sheet	×1
Foam pad, folded blankets, or stacked cardboard — at least 2 inches under you	1 stack
Blankets or quilts	×3
Foil emergency blanket (optional)	×1
Dry hat and dry socks	1 set

1. Tarp down first, as the ground's moisture wall. Pad stack on top of it.
2. One blanket under you on the pad — yes, under — then you, then the rest over you, tucked at the shoulders.
3. Foil blanket, if you have one, goes over the top layer with the shiny side facing you, bouncing your glow back. Leave your face uncovered — foil traps breath moisture.
4. Hat and dry socks on. Loosen everything tight. Eat something first.

FIELD RECIPE · BED WARMER

Makes: a hot-water bottle that pre-heats the bed and holds warmth for hours.

Component	Count
Sturdy bottle that seals — a hot-water bottle, or a hard plastic bottle in a pinch	×1

Hot water — hot tap or heated, never boiling	1 bottleful
Sock, towel, or cloth cover	×1

1. Fill with hot — not boiling — water. Boiling water wrecks seals and scalds.
2. Squeeze the air out, cap it, check the seal upside down over a sink.
3. Cover it with the sock or towel. Straight hot plastic against skin burns — especially skin that can't feel heat well. Always covered, never trapped against bare skin all night.
4. Into the foot of the bed ten minutes before you get in.

3. SHRINK THE SPACE

When the cold gets serious, move the whole household into the smallest defensible room: interior, few windows, one door. Then shrink it again:

- **A tent pitched inside the room** — or a blanket fort over the bed — traps your own body heat in a few cubic feet of air. Campers have slept warm in tents in weather that would kill in a house. The tent doesn't care that it's indoors.
- **Hang blankets over the room's doorways and windows.** Every layer of still air is insulation. A curtained room inside the house can hold ten degrees the rest of the house doesn't have.
- **Live low in winter.** Heat rises — but in a sealed cold house, the floor by the interior wall, away from windows, is where your small heat source does the most good. And sleep off the floor on your pad stack: cold pools at floor level.



START AT THE MIDDLE. WORK OUT ONLY IF YOU'RE RICH IN FUEL.

THE LADDER — EVERY RUNG OUT COSTS TEN TIMES THE FUEL

4. HEATING THE SMALL SPACE

With the space shrunk, modest heat becomes plenty. In order of trust:

- **A wood stove or fireplace with a real chimney** — the only big fire that belongs indoors (Book 3 covers the fire itself; keep its stop signs in this room too).
- **Reflected heat.** A sheet of foil or a shiny baking tray behind the stove bounces glow back into the room instead of into the wall. Heat also bounces off a blanket hung behind *you*.
- **Thermal mass.** Bricks or jugs of water set near the stove drink up heat all evening and hand it back for hours after the fire dies. Free storage, no battery required.
- **Seal the leaks.** A candle flame held near a closed window tells the truth: where it flickers, air is moving, and your heat is leaving. Drafts are winter's broken windows.

FIELD RECIPE · DRAFT STOPPER

Makes: a door snake that kills the cold river that runs under every door.

Component	Count
 Long tube of fabric — a stocking, a sleeve, a sewn pillowcase	×1
 Dry fill — rice, dried beans, sand, or rolled towels	1 fill

1. Fill the tube until it's firm, a little longer than the door is wide.
2. Tie or sew both ends shut.
3. Lay it snug along the door's bottom, on the warm side. One per exterior door — and one on the door of your one warm room.

FIELD RECIPE · WINDOW SKIN

Makes: a clear insulation layer that turns a single cold window into a double one.

Component	Count
Bubble wrap, packing size	1 window's worth
Spray bottle of water	×1

1. Cut the bubble wrap to the glass size, bubbles toward the glass.
2. Mist the glass with water and press the wrap on. The water holds it for months — no tape, no damage.
3. The bubble layer is trapped air, and trapped air is insulation. Light still comes in; a good share of the cold doesn't.

Thick curtains or a hung blanket over the skin handle the night shift.

FIELD RECIPE · CLAY POT HEATER

Makes: a small radiant warmer from candles and flowerpots — a hand-and-lap heater, honestly sized.

Component	Count
Tea light candles	×4
Terracotta pots, two sizes that nest	×2
A non-burnable stand — bricks or a metal tray	×1
A coin or foil square to block the big pot's drain hole	×1

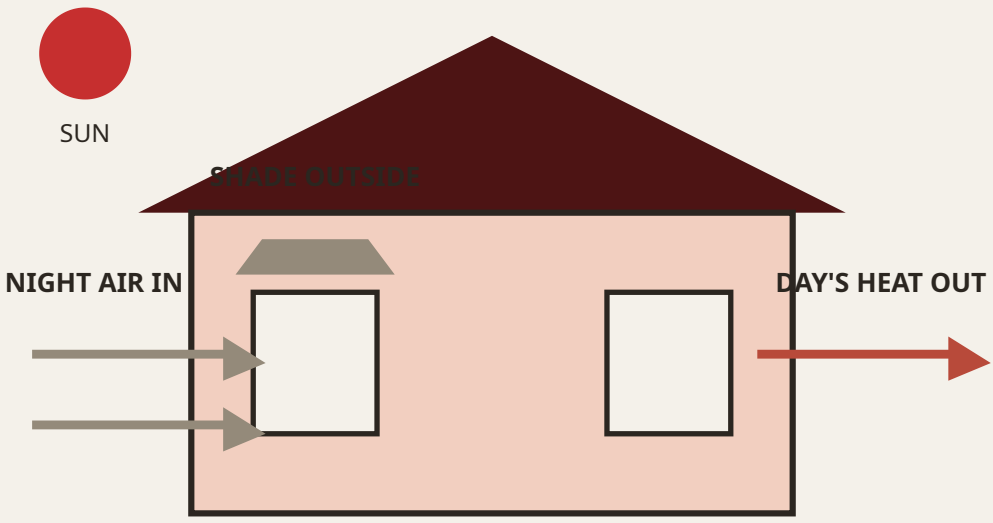
1. Candles on the tray, lit. Small pot upside down over them, its drain hole open — the candles need that air.
2. Big pot upside down over the small one, its drain hole covered. Air flows in low, warms between the pots, and the clay soaks it up.
3. Give it twenty minutes. The pots become a slow, steady radiator about the size of a toaster's warmth.

Honest size: four candles make roughly one person's worth of heat — because that's all a candle is. This warms hands, a lap, a tent's worth of air. It will not heat a room, and anything that burns follows Book 3's rule: never in a sealed space, never while you sleep.

5. COOLING A SPACE

Summer runs the same physics, aimed the other way. The house is a box that collects sun all day — your job is to stop the collecting and dump the heat at night:

- **Shade the glass from outside.** A sheet, awning, or cardboard on the *outside* of a sunny window stops heat before it gets in. Curtains inside are a distant second — the heat is already through the glass.
- **Flush at night.** Open windows on opposite sides of the house after dark and let the night air push the day's heat out — high windows out, low windows in. Then close everything and pull the shades before the sun is up. You're banking coolth like the bricks banked heat.
- **Live low in summer.** The ground floor — the basement if you have one — sits in the earth's own cool. Move the beds down for the hot weeks.
- **Cook outside, or not at all.** Every stove hour is a heater hour. Summer suppers come from the fire book's outdoor chapter now.



SHADE BY DAY · FLUSH BY NIGHT · CLOSE UP AT DAWN

THE HOUSE AS A BATTERY: CHARGE IT WITH COOLTH OVERNIGHT

FIELD RECIPE · SWAMP COOLER

Makes: an evaporative cooler that drops air temperature in dry heat — a fan, a bucket, and water doing the work of an air conditioner.

Component	Count
Bucket or plastic tub with a lid	×1
Small fan — a USB fan off your Book 1 battery rig is perfect	×1
Towel, burlap, or sponge — the wick	×1
Water	1 fill

1. Cut a hole in the lid for the fan, blowing down into the bucket, and a ring of vent holes around the bucket's upper sides.
2. Line the inside walls with the towel, its bottom edge standing in the water. The towel wicks water up and stays wet.
3. Fan on. Air is pushed past the wet towel, water evaporates into it, and the air that leaves the vents is measurably cooler.
4. Top up the water. Evaporation is the fuel — a bucketful a day in real heat.

Honest size: evaporation only works when the air is dry enough to drink. In humid heat this becomes a fan with a hobby — plain moving air and wet skin (Entry 6) are your coolers there.

6. COOLING YOU

The space can only do so much. The person is the real target, same as winter:

- **Wet the pulse points.** A wet cloth on the neck, wrists, armpits, and groin cools the blood near the surface — the body's radiator hoses. Rewet as it dries; the evaporation is the cooling.
- **Soak the shirt.** A wrung-out wet shirt plus any breeze is a swamp cooler you wear. In dry heat it can make the difference between functioning and folding.
- **Drink on schedule, not on thirst.** Thirst runs late in heat. Water rules are Book 2's — but in a heat wave, drink before you're thirsty and rest before you're done. Work at dawn and dusk; shade at noon is a job too.
- **Loose, light, covered.** Desert peoples wrap up for a reason: light cloth shades skin, holds a layer of cooler air, and slows the sweat you can't afford to lose.

7. WHAT TO SCAVENGE FOR WARM AND COOL

Insulation is the most scavengable thing in the world — it's bulky, common, and nobody else is fighting over it:

- **Cardboard.** The poor man's foam pad, and genuinely good: corrugated air pockets insulate floors, beds, walls, and boots (a cardboard insole is a real thing). Stack it, keep it dry.
- **Blankets, quilts, sleeping bags, curtains.** Loft is warmth — judge by puffiness per weight, not by looks. Wool keeps some warmth even damp; cotton gives up when wet; synthetics sit between.
- **Foam anything** — pads, cushions, pool noodles, packing foam. Foam doesn't crush flat, which makes it the under-you champion.
- **Bubble wrap and plastic sheeting** — window skins, moisture walls, wind shells. A big clear sheet is also a solar still's cousin and a rain collector's friend (Book 2 sends its regards).
- **Foil emergency blankets** — near-weightless, and they bounce glow both ways: around you in winter, over a window or a roof in summer.
- **Clay pots, bricks, jugs** — thermal mass and heater bodies (Entry 4). Heavy junk nobody loots is still junk that works.
- **Fans, any size that spins** — a USB fan on a battery bank (Book 1) is summer's best friend, and the swamp cooler's heart.

> **Judging it all:** dry beats thick. A thin dry blanket outperforms a fat wet one every night of the week — water conducts heat away 25 times faster than air, and wet insulation is just a cold water jacket. If it isn't dry, dry it first (Book 3's fire earns its keep again), or leave it.

8. STOP SIGNS

- **The "umbles."** Hypothermia doesn't announce itself — it arrives as stumbles, mumbles, fumbles, and grumbles, and the person having it is the last to know. Umbles in a cold companion means stop, shelter, dry layers, warm sweet drinks, skin-to-skin warmth in the cocoon. No alcohol — it feels warm and steals heat; a liar in a bottle.
- **Shivering that stops.** That's not improvement, that's the furnace running out of fuel. Emergency, not progress.
- **Frostbite is pale, hard, and numb.** Rewarm slowly in warm — not hot — water. Never rub it, never use dry fire heat. And never rewarm a foot if it might refreeze on the walk home; walking on a frozen foot beats thawing and refreezing it.
- **Heat exhaustion vs. heat stroke.** Pale, sweaty, dizzy, nauseous: heat exhaustion — shade, rest, water, wet cloths. But hot, dry, and confused is heat *stroke*: the thermostat is broken and the person is cooking. Cool them fast by any means you have — wet the whole body, fan hard — and treat it as the emergency it is.
- **Carbon monoxide, again.** Every heater that burns follows the fire book's law: exhaust up and away, never a sealed room, never while everyone sleeps. Headache near a heater means air, now.
- **Hot bottles and numb skin.** Skin that can't feel heat can't warn you. Bed warmers stay covered, warm — never scalding — and never pinned under a sleeping body.

9. THE WHOLE BOOK ON ONE CARD

- Warm the person, then the bed, then one room. The house is the last resort, not the plan.
- Heat leaves four ways: touch, wind, glow, sweat. Plug them in that order.
- Insulation under you beats blankets over you. The ground steals all night.
- Loft is warmth. Wet insulation is a cold water jacket — dry beats thick.
- Shrink the space: one room, a tent inside it, blankets on the gaps.
- Bricks and water jugs bank a stove's heat and pay it back for hours.
- Summer mirrors winter: shade the glass from outside, flush the house at night, close up at dawn.
- A swamp cooler cools dry heat; wet skin and moving air cool the rest.
- Umbles mean hypothermia. Hot, dry, and confused means heat stroke — cool fast.
- Every burning heater obeys the fire book: vented, watched, never sealed in.

ALL BOOKS, A–Z

THE MILLENNIAL'S GUIDE TO THE AFTERMATH — STAYING WARM AND KEEPING COOL, EXPLAINED